

Key features of parabolas



- 1
 - a Downwards
 - b Upwards
- 2 Upward
- 3 No, because the entire parabola will be above the x -axis. Also, there is no solution for the equation $x^2 + 6 = 0$.
- 4
 - a No
 - b Yes
 - c No
 - d Yes
- 5
 - a $x = 0, 4$
 - b $y = 0$
 - c 4
- 6
 - a Concave up
 - b $y = 9$
 - c $y = 1$
 - d $x = -2$
 - e $x < -2$
- 7
 - a $(-2, 0)$
 - b Vertex: $(-2, 0)$
 - c
 - i False
 - ii True
 - iii True
 - iv True
- 8
 - a 2
 - b 1
- 9 Concave up
- 10
 - a $(-4, 2)$
 - b 2
- 11 $x = 3$
- 12
 - a No
 - b No
 - c Yes
- 13 Quadratic function A



15 $y = 25$

16

a $x = 14, 6$

b $x = 10$

c Concave down

d $y = 16$

17 The x -coordinate of the vertex of the parabola occurs at $x = \frac{-b}{2a}$. The y -coordinate of the vertex is found by substituting $x = \frac{-b}{2a}$ into the parabola's equation and evaluating the function at this value of x .

18 $x = \frac{1}{4}$

19 $x = -3, 2$

20

a False

b True

21 $x = \frac{-3 \pm \sqrt{93}}{6}$

Graphing quadratics

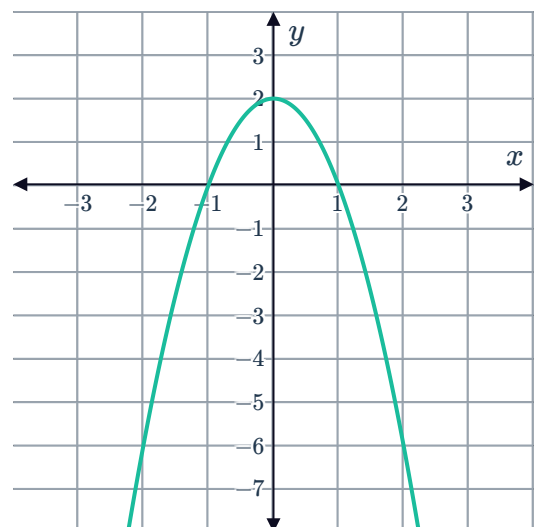
22

a Concave down

c $(0, 2)$

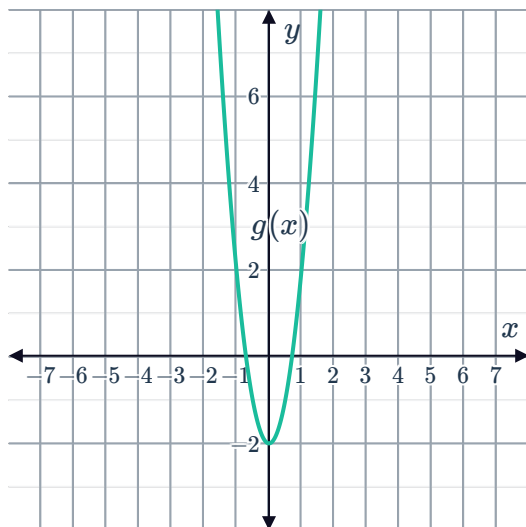
b More steep

d

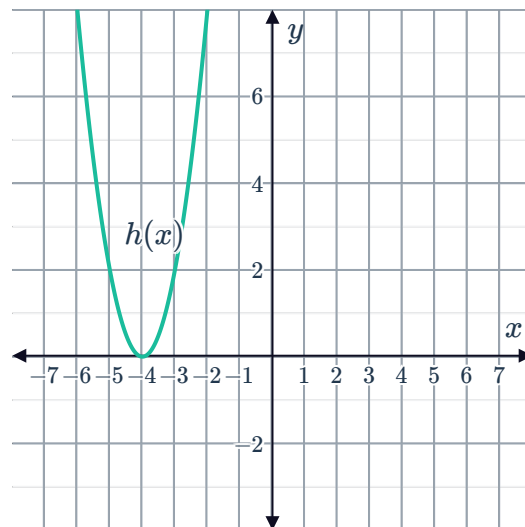


23

a



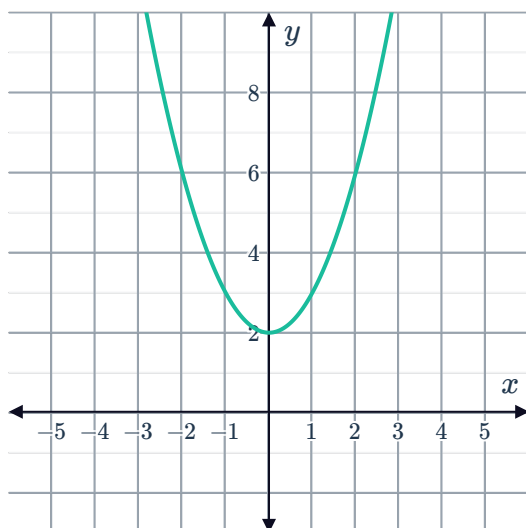
b



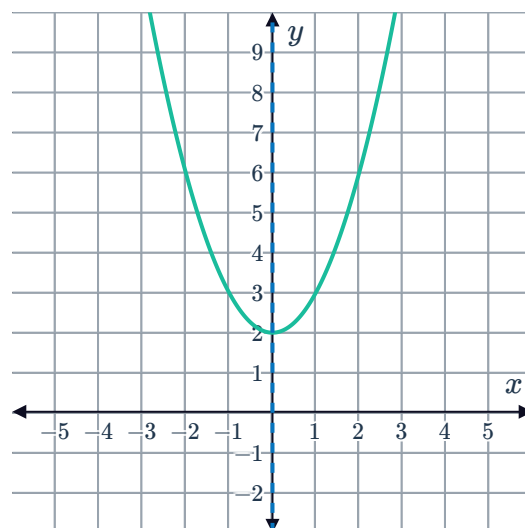
24 $y = -x^2 + 5$

25

a



b



c $(0, 2)$

26

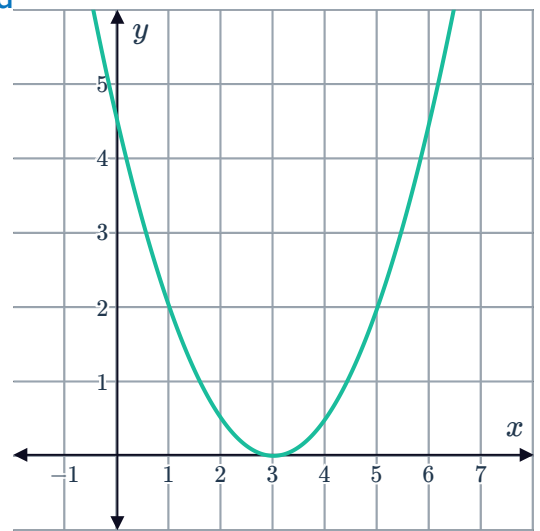
a Concave up

b Less steep

c $(3, 0)$



d



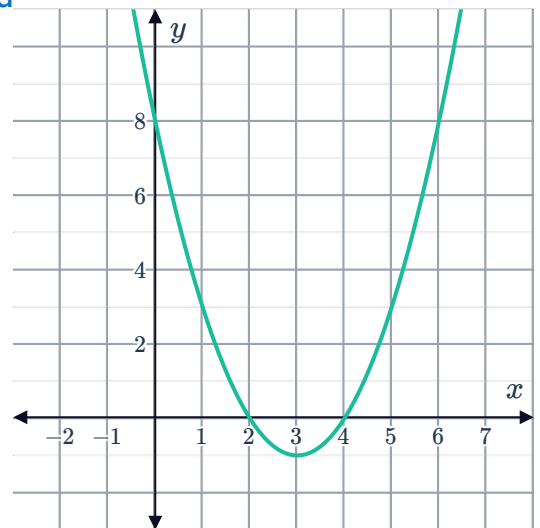
27

a $x = 4, 2$

b $y = 8$

c $(3, -1)$

d



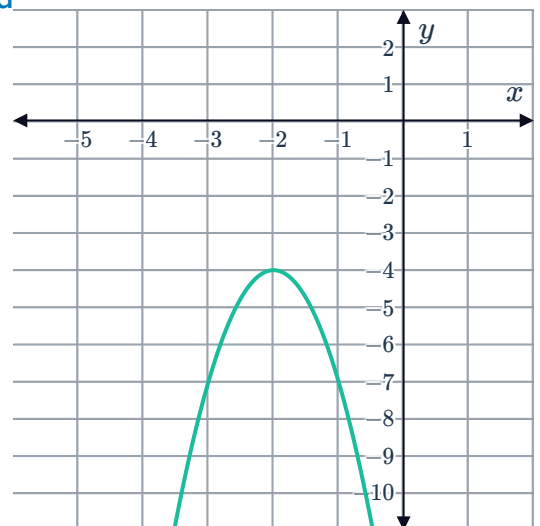
28

a $(-2, -4)$

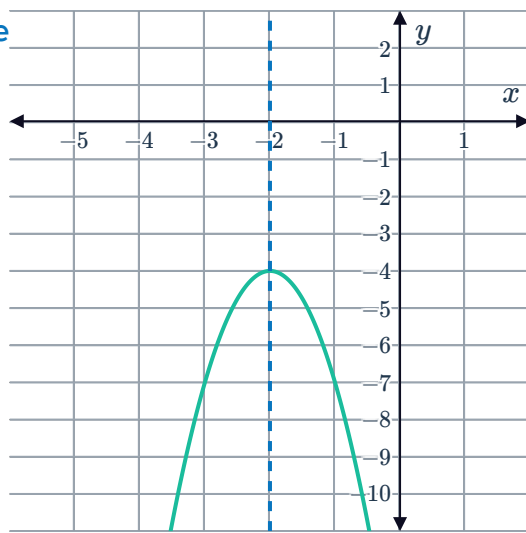
b $x = -2$

c $y = -7$

d

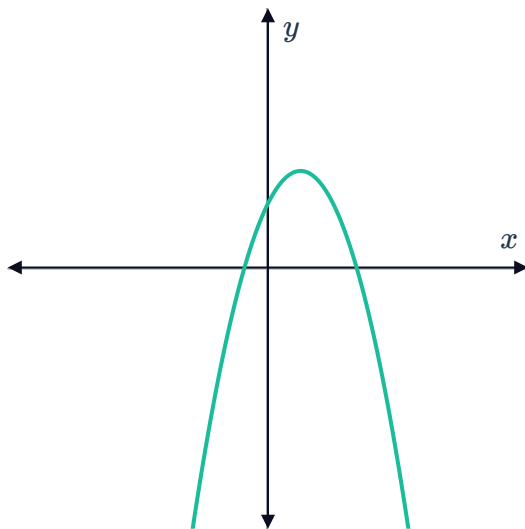


e

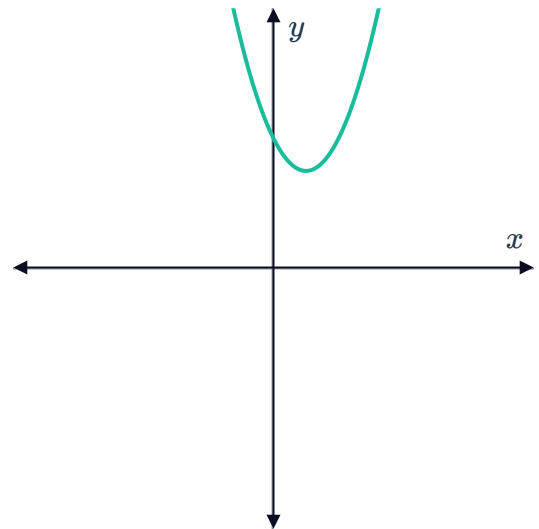


29

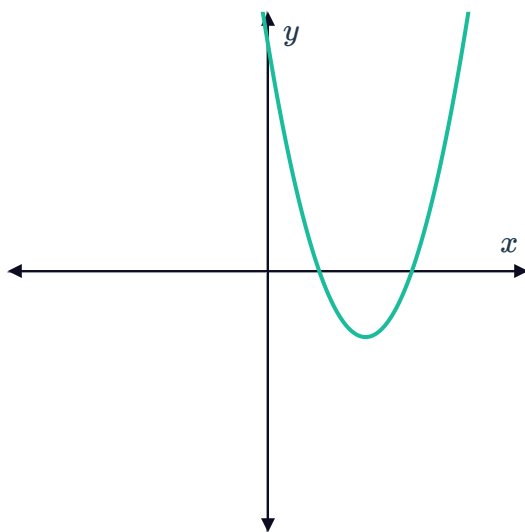
a



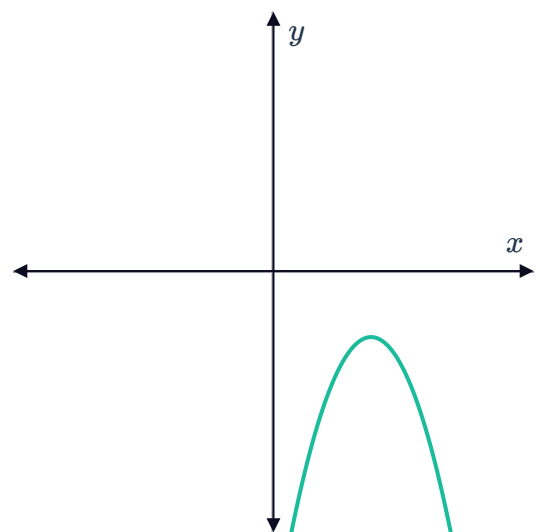
b



c



d



30

a 8

c

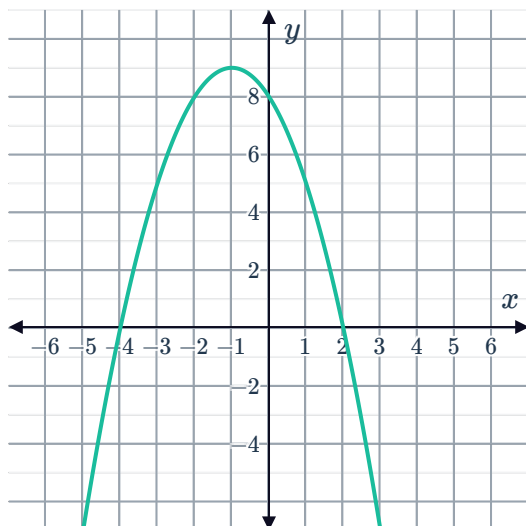
x	-5	-3	-1	1	3
y	-7	5	9	5	-7



b $x = 2, -4$

d $(-1, 9)$

e



31

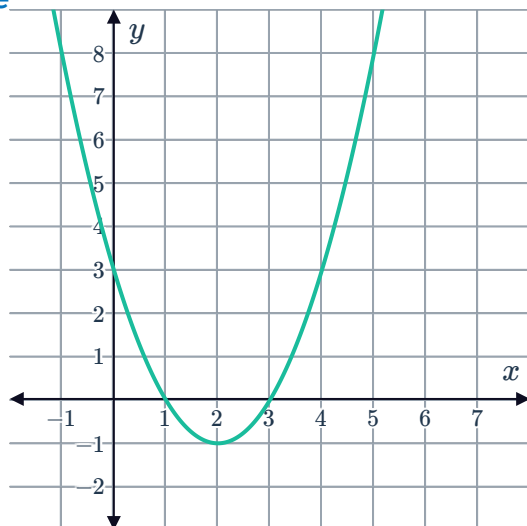
a $y = 3$

b $x = 3, 1$

c $x = 2$

d $(2, -1)$

e



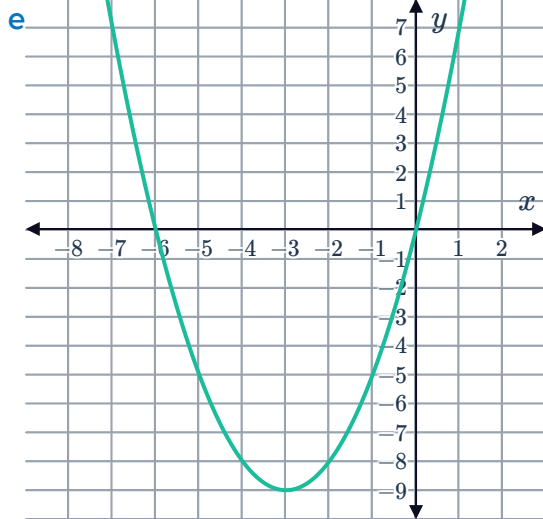
32

a $y = 0$

b $x = 0, -6$

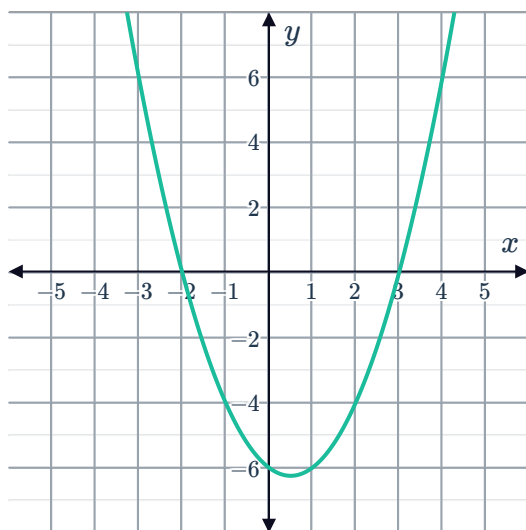
c $x = -3$

d $(-3, -9)$

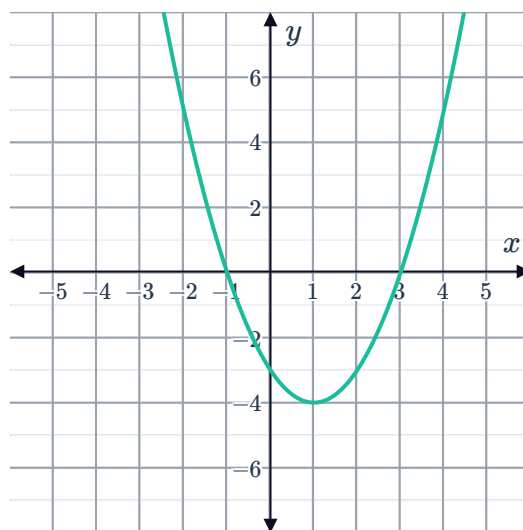


33

a

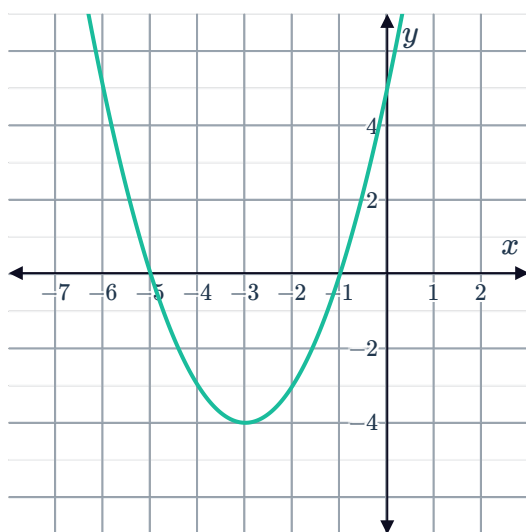


b

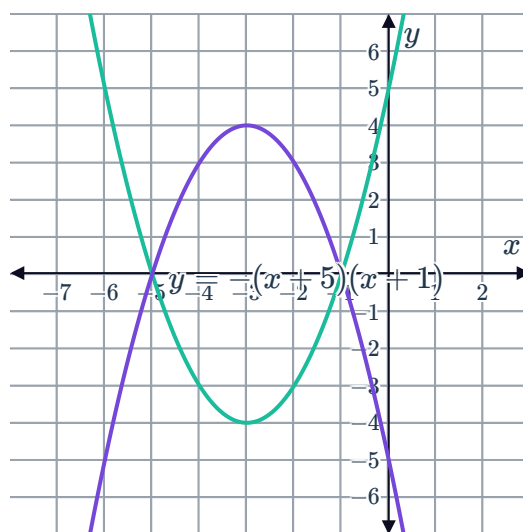


34

a



b



35

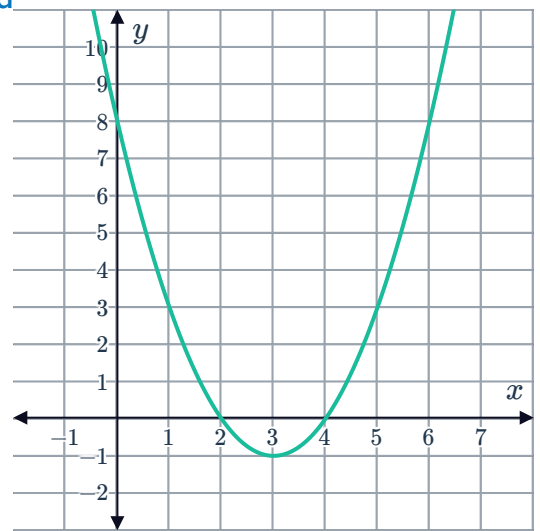
a $(x - 2)(x - 4)$

b $x = 2, 4$

c $(3, -1)$



d



36

a $y = (x + 4)(x - 3)$

c $y = -12$

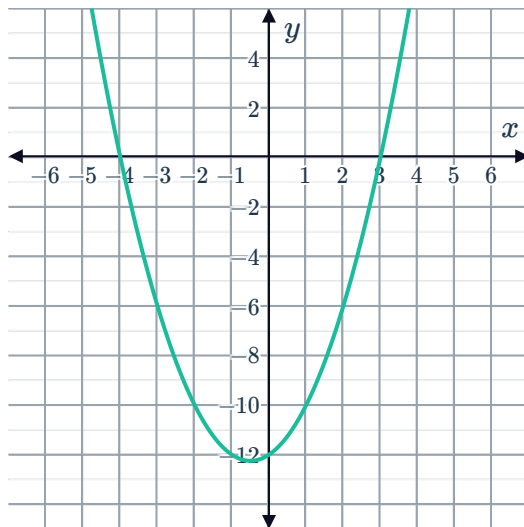
e $\left(-\frac{1}{2}, -\frac{49}{4}\right)$

b $x = -4, 3$

d $x = -\frac{1}{2}$

f Concave up

g



37

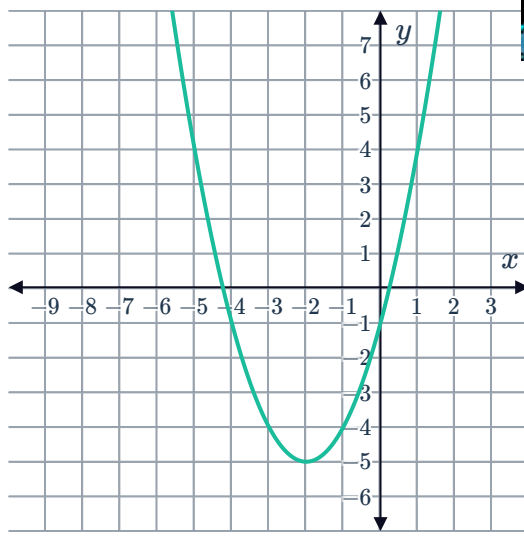
a $y = (x + 2)^2 - 5$

c $(-2, -5)$

b $y = -1$

d Concave up

e



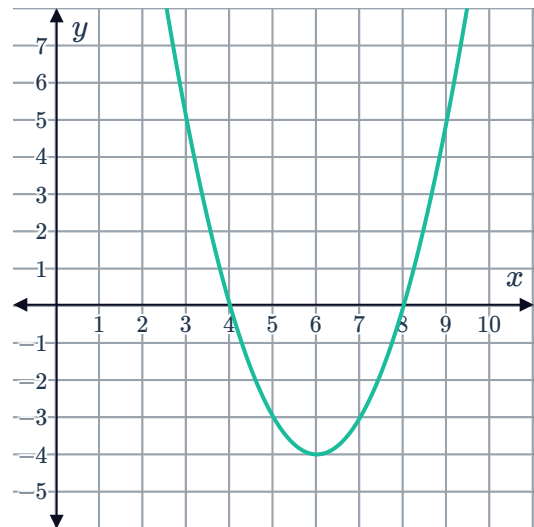
38

a $x = 8, 4$

c $(6, -4)$

b $y = (x - 6)^2 - 4$

d

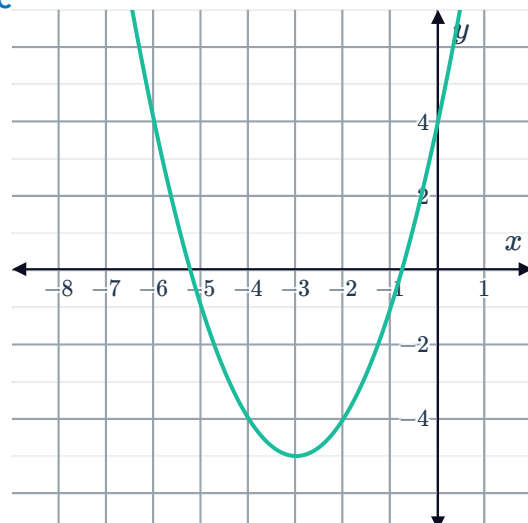


39

a $x = -3$

b -5

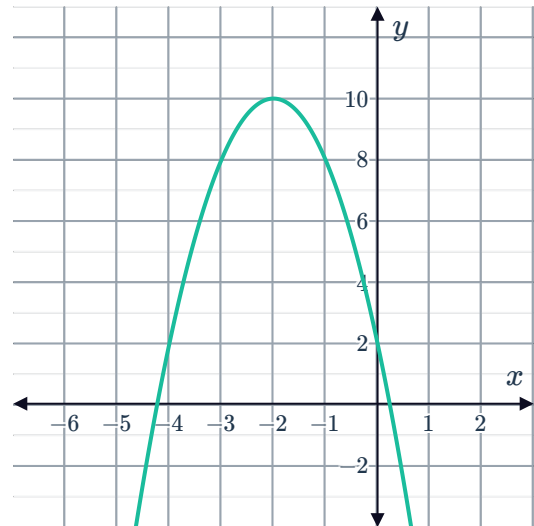
c



40 a $(-2, 10)$



b

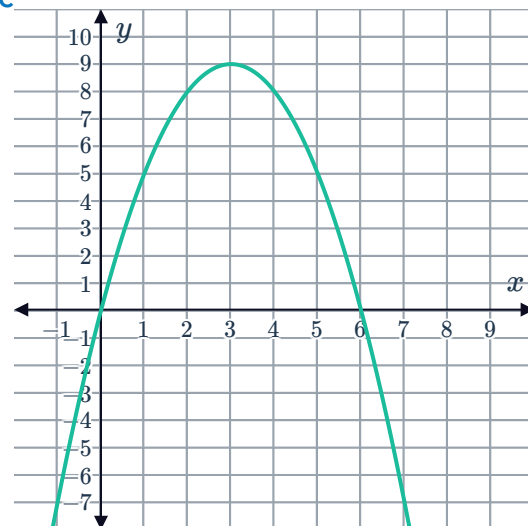


41

a $x = 0,6$

b $(3, 9)$

c



42

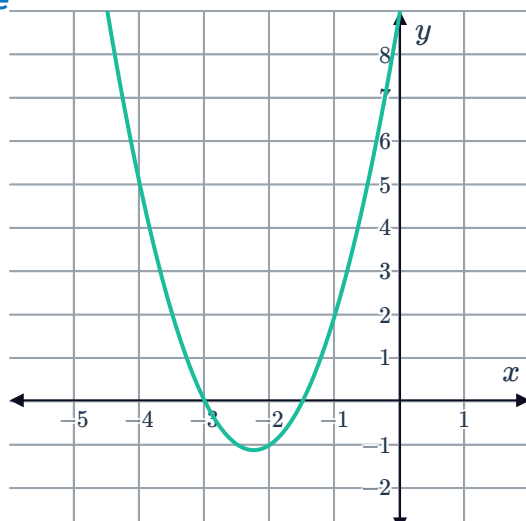
a $x = -\frac{3}{2}, -3$

b $y = 9$

c $x = -\frac{9}{4}$

d $y = -\frac{9}{8}$

e





Graphing quadratics using technology

43

a

i $(8, 7)$

ii $(0, 263)$

b

i $(-6, 4)$

ii $(0, -140)$

44

a

$(0, -12)$

b

No

c

$x < 0$

45

a

$x = -3.1$

b

-16.61

46

a

$x = 2, 6$

b

$y = -24$

c

$x = 4$

d

$y = 8$

47

a

$x = 1.55$

b

$y = 2.95$

c

Yes

d

$x = 3.58, -0.47$

48

a

$x = 2.75$

b

$y = -9.10$

c

Yes

d

$x = -0.42, 5.91$